

Method of Variation of Parameters (Rachel Batra)

Ques: Solve $y'' - y = x$

Sol: The general solution comprises of two parts: $y = y_h + y_p$

For y_h : We solve the homogeneous part using ode2.

```
--> ode2('diff(y,x,2) - y = 0', y, x);
```

$$y = k_1 e^x + k_2 e^{-x}$$

```
-- y1: rhs(%); /*giving a name to the RHS of previous
> output*/
```

$$k_1 e^x + k_2 e^{-x}$$

```
--> y1: %e^x; y2: %e^(-x); r: x; /*assigning relevant names*/
```

$$e^x$$

$$e^{-x}$$

$$x$$

For y_p : We calculate the Wronskian of y_1 and y_2

```
-->
-- A: matrix (
> [y1, y2],
  [diff(y1, x), diff(y2, x)] /*Shortcut for this command: Matrix -
  > Enter Matrix*/
);
```

$$\begin{pmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{pmatrix}$$

```
--> W: determinant(A);
```

$$-2$$

We now find u_1 and u_2

```
-- u1: integrate(-y2 * r / W, x); /*Shortcut for this command: Calculus ->
> Integrate*/;
```

$$\frac{(-x - 1)e^{-x}}{2}$$

```
--> u2: integrate(y1 * r / W, x);
```

$$-\frac{(x - 1)e^x}{2}$$

Now, $y_p = u_1 y_1 + u_2 y_2$

--> `yp : ratsimp ( u1 · y1 + u2 · y2 ) ;` /*This command simplifies the expression*/

$$-x$$

The general solution is

--> `'y = yh + yp ;`

$$y = \%k1 \%e^x + \%k2 \%e^{-x} - x$$

Ques. $y'' + y = \sec x$

--> `ode2 ( 'diff ( y , x , 2 ) + y = 0 , y , x ) ;`

$$y = \%k1 \sin (x) + \%k2 \cos (x)$$

--> `yh : rhs ( % ) ;`

$$\%k1 \sin (x) + \%k2 \cos (x)$$

--> `y1 : sin ( x ) ; y2 : cos ( x ) ; r : sec ( x ) ;`

$$\sin (x)$$

$$\cos (x)$$

$$\sec (x)$$

--> `A : matrix (`
`[ y1 , y2 ],`
`[ diff ( y1 , x ) , diff ( y2 , x ) ]`
`);`

$$\begin{pmatrix} \sin (x) & \cos (x) \\ \cos (x) & -\sin (x) \end{pmatrix}$$

--> `W : determinant ( A ) ;`

$$-\sin (x)^2 - \cos (x)^2$$

--> `u1 : integrate ( - y2 · r / W , x ) ;`

$$x$$

--> `u2 : integrate ( - y1 · r / W , x ) ;`

$$-\frac{\log \left(1 - \sin (x)^2\right)}{2}$$

--> `yp : ratsimp ( u1 · y1 + u2 · y2 ) ;`

$$-\frac{\cos(x) \log(1 - \sin(x)^2) - 2x \sin(x)}{2}$$

--> 'y = yh + yp;

$$y = -\frac{\cos(x) \log(1 - \sin(x)^2) - 2x \sin(x)}{2} + \%k1 \sin(x) + \%k2 \cos(x)$$

Ques.y''-2y'+y=(e^x)/x

-- ode2('diff(y,x,2)-2*(diff(y,x))+y=0,y,x); /* Use 'diff' when u want dy/dx to be
> shown and not the value of the differential*/

$$y = (\%k2x + \%k1)\%e^x$$

--> yh:rhs(%);

$$(\%k2x + \%k1)\%e^x$$

--> yh=y1:x*(%e^x);y2:%e^x;r:(%e^x)/x;

$$(\%k2x + \%k1)\%e^x = x\%e^x$$

$$\%e^x$$

$$\frac{\%e^x}{x}$$

-->

--> A:matrix(
[y1,y2],
[diff(y1,x),diff(y2,x)]
);

$$\begin{pmatrix} x\%e^x & \%e^x \\ x\%e^x + \%e^x & \%e^x \end{pmatrix}$$

--> W:determinant(A);

$$x\%e^{2x} - \%e^x(x\%e^x + \%e^x)$$

--> u1:integrate(-y2*r/W,x);

$$\log(x)$$

--> u2:integrate(y1*r/W,x);

$$-x$$

--> yp:ratsimp(u1*y1+u2*y2);

$$x\%e^x \log(x) - x\%e^x$$

```
--> 'y = yh + yp ;
```

$$y = x e^x \log(x) + (\%k2x + \%k1) e^x - x e^x$$

Created with [wxMaxima](#).

The source of this Maxima session can be downloaded [here](#).